

Yonatan Tesfagiorgis

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EDUCATION

University of Houston, Houston, TX

Expected May 2026

Bachelor of Science, Computer Science

Texas State Technical College, Rosenberg, TX

July 2019

Associate of Science, Cybersecurity

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, PHP, MySQL, HTML, CSS, TCP/IP

Frameworks & Libraries: React.js, Vite, Node.js, Express.js, Flask, scikit-learn, PyTorch, NLTK, spaCy, TensorFlow

Web Development: RESTful API Development, Frontend & Backend Development, Git, GitHub

AI/ML: LLMs, Retrieval-Augmented Generation (RAG), Prompt Engineering, Transformers, BERT,

Hugging Face, LangChain, Vector Databases (FAISS), Docker

Data Science: Pandas, NumPy, Matplotlib, R, EDA, Model Training & Evaluation, Logistic Regression, Decision Trees, SVM,

CNN, RNN, Reinforcement Learning, LSTM, Autoencoders, Diffusion Models

PROJECT EXPERIENCE

GitHub: <https://github.com/ravenfire24>

University of Houston

NLP Aspect-Based Sentiment Retrieval System (Python, scikit-learn, NLTK, BERT)

August 2025 – November 2025

- Developed an NLP system processing 10,000+ Amazon reviews for aspect-based sentiment classification.
- Integrated Logistic Regression and BERT embeddings to improve sentiment classification accuracy.
- Implemented semantic similarity search using sentence-transformers and cosine similarity.
- Increased precision across aspect queries compared to keyword-based approaches.

AI-Powered Document Assistant (RAG + LLMs) (LangChain, Hugging Face, FAISS, Streamlit, Python)

January 2026 – March 2026

- Designed and deployed a Retrieval-Augmented Generation (RAG) system using LangChain and Hugging Face LLMs to enable conversational querying over PDF documents.
- Built a semantic search pipeline using sentence-transformer embeddings and a FAISS vector database for efficient and accurate document retrieval.
- Applied prompt engineering and context-aware retrieval techniques to improve response relevance and reduce hallucinations in LLM outputs.
- Developed and deployed a real-time Streamlit chat interface on Hugging Face Spaces for scalable and interactive document exploration.

Volunteer Management Application (Full-Stack: Python/Flask, React, MySQL, ML)

June 2025 - August 2025

- Built scalable RESTful APIs for authentication, event scheduling, and user management.
- Developed a responsive React dashboard with real-time analytics and visualization.
- Automated report generation (PDF/CSV) and tracked participation across 50+ events.
- Enabled data-driven decision-making using analytics insights.

Chest X-Ray Pneumonia Detection (Python, PyTorch, DenseNet121)

January 2026 – March 2026

- Built a CNN to classify chest X-rays as normal, pneumonia, or tuberculosis using 15,000+ images with augmentation.
- Fine-tuned DenseNet121 with class-weighted loss, achieving 85%+ validation accuracy.
- Visualized results with confusion matrices and saved the model for deployment in clinical workflows.

WORK EXPERIENCE

Production Assembly Operator – Randstad

September 2020 – December 2021

- Supervised 5 – 7 production line workers and optimized material flow, improving workflow efficiency by 15% and ensuring on-time delivery of products.
- Trained Staff in safety and quality standards, reducing incidents by 25 %.
- Operated and maintained machines, reducing downtime incidents by 20 %

LEADERSHIP EXPERIENCE

University of Houston - Cougar CS Organization

January 2024 – Present

- Led C++ workshops and collaborative problem-solving sessions for 50+ students.